E-Commerce Database System

Relational Schema Architecture, 3NF Normalization & Live Implementation

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Course: CSE311 — Database Management Systems (DBMS)

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Catalog Cart Orders Dashboard Rahim Uddin

Catalog

4 items on the shelf

Search products... | All categories

SKU-MOUSE-01
Wireless Mouse
Computer Accessories
$12.50 Add to cart

SKU-KEYB-01
Mechanical Keyboard
Computer Accessories
$45.00 Add to cart

SKU-SPKR-01
Bluetooth Speaker
Electronics
$30.00 Add to cart

SKU-BOOK-01
Database Systems Textbook
Books
$25.00 Add to cart

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Catalog Cart Orders Dashboard Ayesha Khan

Order history

ORDER	DATE	STATUS	TOTAL	
#5	8/19/2026	PENDING	$12.50	View →

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Catalog Cart Orders Dashboard Jane Doe

Your cart

Shopping as Jane Doe

Wireless Mouse

$12.50 each

− 1 + Remove

ORDER SUMMARY

Wireless Mouse ×1	$12.50
Total	**$12.50**

Ship to

12 Elm St, Dhaka

Payment method

Card

Place order

Order history

ORDER	DATE	STATUS	TOTAL	
#5	8/19/2026	PENDING	$12.50	View →

01. OVERVIEW

Introduction & Project Motivation

System Purpose

- Design a robust OLTP relational database for e-commerce transactions.
- Manage user credentials, recursive categories, dynamic shopping carts, and orders.
- Backend designed for CSE311 live project integration.

Core Challenges

- Preventing data redundancy across multi-vendor inventory.
- Preserving historical financial records against future price changes.
- Enforcing strict referential integrity across 12 distinct tables.

Engineering Goals

- 100% compliance with Third Normal Form (3NF / BCNF).
- ACID transactional consistency for cart and order checkout.
- Live analytical query reporting for store managers.

02. THE PROBLEM

Problem Statement: Flaws in Flat/Unnormalized Data

Insertion Anomalies

- Cannot add a new product (e.g. Wireless Mouse) without an active order.
- Cannot onboard new suppliers (e.g. BookHouse BD) without orders.
- Forces dummy NULL values into database.

Update Anomalies

- Updating customer address requires changes in dozens of order rows.
- Changing a supplier cost or catalog price risks corrupting past sales.
- Massive operational overhead & data conflicts.

Deletion Anomalies

- Deleting a cancelled purchase order accidentally erases user accounts.
- Purging old orders can wipe product catalog records entirely.
- Cascading loss of critical business data.

03. NORMALIZATION

Database Normalization Walkthrough: UNF to 3NF

Form	Condition & Transformation	Status / Remaining Issue
UNF	Single flat table with repeating groups in cells like '(Mouse, 2, $10), (Keyboard, 1, $25)'	Violates 1NF; non-atomic data
1NF	Split multi-item cells into individual rows with atomic values	Partial dependency: cust_name depends only on order_id
2NF	Removed partial dependencies by splitting into Order_Tbl and Order_Item	Transitive dependency: address depends on customer
3NF / BCNF	Removed transitive dependencies by separating Customer and Address tables	Full 3NF target achieved; zero data anomalies

Why 3NF / BCNF is the Optimal Standard

- Eliminates all update, insert, and delete anomalies while keeping query joins fast and predictable.
- Avoids over-engineering (no 4NF/5NF multi-valued dependencies exist in standard e-commerce models).

04. RELATIONAL SCHEMA (PART 1)

Database Tables: User, Catalog & Sourcing Entities

Table Name	Primary Key	Foreign Keys (FK) & Target	Key Attributes & Types
customer	customer_id	—	first_name, last_name, email, phone, password_hash, created_at
address	address_id	customer_id -> customer.customer_id	address_line, city, state, zip_code, country, is_default
category	category_id	parent_category_id -> category.category_id	category_name, parent_category_id (Recursive Tree)
supplier	supplier_id	—	supplier_name, contact_email, phone
product	product_id	category_id -> category.category_id	product_name, sku, description, price, stock_quantity, created_at
product_supplier	(product_id, supplier_id)	product_id -> product, supplier_id -> supplier	cost_price decimal(10,2) [M:N Junction Table]

05. RELATIONAL SCHEMA (PART 2)

Database Tables: Cart, Orders, Payments & Reviews

Table Name	Primary Key	Foreign Keys (FK) & Target	Key Attributes & Types
cart	cart_id	customer_id -> customer.customer_id	created_at (Session persistence per user)
cart_item	cart_item_id	cart_id -> cart, product_id -> product	quantity int(11)
order_tbl	order_id	customer_id -> customer, address_id -> address	order_date, status ENUM('PENDING', 'SHIPPED', ...), total_amount
order_item	order_item_id	order_id -> order_tbl, product_id -> product	quantity, unit_price decimal(10,2) [Price Snapshot]
payment	payment_id	order_id -> order_tbl.order_id	payment_method ENUM('CARD','COD',...), amount, status ENUM
review	review_id	product_id -> product, customer_id -> customer	rating tinyint(4), comment text, review_date

06. TECHNICAL HIGHLIGHTS

Key Technical Design Highlights

Price Snapshotting

- order_item stores unit_price as decimal(10,2) at checkout.
- Guarantees future catalog price changes never alter past receipts.
- Essential for legal audits and accounting compliance.

Multi-Payment Architecture

- 1:M relationship between order_tbl and payment.
- Supports CARD, MOBILE BANKING, COD, and WALLET.
- Allows partial, split, or multi-tender settlement.

Recursive Categories & Rules

- Self-referencing parent_category_id FK.
- Enables unlimited category depth (Electronics > PC > Mouse).
- ON UPDATE RESTRICT & ON DELETE CASCADE enforce strict integrity.

07. FRONTEND IMPLEMENTATION

Application Showcase: Catalog, Cart & Orders

Live Product Catalog (Catalog View)

- Integrated with dynamic SQL queries pulling from product & category.
- Sample Products displayed: Wireless Mouse ($12.50), Mechanical Keyboard ($45.00), Bluetooth Speaker ($30.00), Database Systems Textbook ($25.00).
- Keyword search and category dropdown filter normalized tables in real-time.
- Add-to-cart inserts/updates into cart and cart_item tables.

Order History & Checkout Flow

- Checkout moves active cart_items into immutable order_item rows.
- Live Order Tracking: Order #5 | Date: 8/19/2026 | Total: $12.50 | Status: PENDING.
- User profile session switching (e.g., Rahim Uddin, Ayesha Khan).
- Foreign key constraints guarantee orders attach to verified customer addresses.

08. STORE METRICS & QUERIES

Application Showcase: Store Analytics Dashboard

Dashboard Metric Box	SQL Underlying Query / Tables	Live Project Values (Screenshot Data)
Top Spenders	SUM(total_amount) JOIN customer GROUP BY customer_id	Jane Doe ($70.00)
Best Sellers	SUM(quantity) JOIN product GROUP BY product_id	Textbook (10 sold), Mouse (4 sold), Keyboard (2 sold)
Product Ratings	AVG(rating), COUNT(*) FROM review GROUP BY product_id	Wireless Mouse (5★), Mechanical Keyboard (4★)
Low Stock (< 50)	SELECT product_name, stock_quantity WHERE stock < 50	Database Systems Textbook (30 left)
Inactive Customers	customer LEFT JOIN order_tbl WHERE order_id IS NULL	Rahim Uddin (rahim.uddin@mail.com)
Payment Summary	COUNT(payment_id), SUM(amount) WHERE status='SUCCESS'	5 Successful Payments ($390.00 Total)
Supplier Cost List	SELECT supplier_name, product_name, cost_price	BookHouse BD ($15.00), TechSource Ltd ($8.00 / $30.00)

09. CONCLUSION

Summary & Key Project Takeaways

3NF & BCNF Certified

- Eliminated all insertion, update, and deletion anomalies.
- 12 fully normalized tables connected via strict FK constraints.
- Zero redundant records; guaranteed ACID compliance.

Full Stack Tested

- Connected seamlessly with the CSE311 project web interface.
- Supports end-to-end shopping: Catalog -> Cart -> Order -> Payment -> Review.
- Accurate historical tracking via price snapshotting.

Student Info

- Name: Sharfin Zaman Swaccha
- Student ID: 2323713042
- Course: CSE311 (Database Management Systems)
- Solid schema modeling ensures scalable database performance.